

DETERMINISTIC PATH ENSEMBLES, DRIVEN COLLISION MAPS, AND ENDOGENOUS CONJUGATE PARAMETERS

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ABSTRACT. We separate the physical generalized-baker impact system from its coupled symbolic natural extension. The physical branches are realized by exact autonomous Hamiltonian channels with a canonical work pair, while the symbolic map is the correctly coupled update $(x, y) \mapsto (\sigma x, x_0 y)$. A geometric mass norm on compatible bulk, face, corner, and higher flag currents is nondegenerate under arbitrary refinement and supports every finite material-response order. The response theorem explicitly includes the transported-observable convention and reduces to zero for a genuinely fixed physical observable.

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1. ROUND-FOURTEEN POSITIVE CLOSURE: PHYSICAL IMPACT SUSPENSION AND A NONDEGENERATE FLAG-CURRENT COMPLEX

The physical section and the symbolic natural extension are different objects. The section is the unit square with its ordinary area form and with the seam orbits removed; the two-sided shift is only a coding extension. This distinction removes the inverse-limit dimension error. The response space

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below is based on geometric mass on the fixed square, rather than an infimum of cellwise norms with a depth-decaying factor.

1.1. The coupled natural extension. Let $I_i(a) = (s_{i-1}(a), s_i(a))$, with width $w_i(a)$, and put

$$B_a(q, p) = \left(\frac{q - s_{i-1}(a)}{w_i(a)}, s_{i-1}(a) + w_i(a)p \right), \quad q \in I_i(a).$$

Let $\Sigma^+ = \{1, 2, 3, 4\}^{\mathbb{N}_0}$ and $\Sigma^- = \{1, 2, 3, 4\}^{\mathbb{N}}$. For $x = (x_0, x_1, \dots)$ and $y = (y_1, y_2, \dots)$, set

$$\mathcal{T}(x, y) = (\sigma x, x_0 y), \quad x_0 y = (x_0, y_1, y_2, \dots).$$

Theorem 1.1 (Physical coding and the coupled extension). *Let $\mathcal{S}_a$ be the union of all forward and backward iterates of the vertical and horizontal seams, and put $S_a^\circ = (0, 1)^2 \setminus \mathcal{S}_a$. There is a bimeasurable coding*

$$\mathbf{c}_a : S_a^\circ \longrightarrow \Sigma^+ \times \Sigma^-$$

onto a full Bernoulli-measure set such that

$$\mathbf{c}_a B_a = \mathcal{T} \mathbf{c}_a, \quad (\mathbf{c}_a)_\#(dq dp) = \nu_a^+ \otimes \nu_a^-.$$

In particular the newly created past symbol is the present future symbol; there is no inverse of an independent one-sided past shift.

Proof. For a finite future word $i_0 \dots i_n$, the corresponding horizontal coordinate interval has length $\prod_{k=0}^n w_{i_k}(a)$. For a finite past word $j_1 \dots j_m$, the vertical coordinate interval has length $\prod_{\ell=1}^m w_{j_\ell}(a)$. Their product rectangles are nested and their diameters tend to zero. Hence every two-sided word outside the endpoint ambiguities determines one point, and every point outside $\mathcal{S}_a$ determines one word. The rectangle areas factor into the Bernoulli weights. Applying B_a deletes i_0 from the future and prepends it to the past, which is exactly $\mathcal{T}$. $\square$

1.2. An autonomous Hamiltonian impact suspension. Write $C_i = I_i(a) \times (0, 1)$. On C_i the branch $B_{a,i}$ is an exact symplectic diffeomorphism from the vertical rectangle C_i to the horizontal rectangle $(0, 1) \times I_i(a)$. Its primitive difference is

$$B_{a,i}^*(p dq) - p dq = dG_{a,i}, \quad G_{a,i}(q, p) = \frac{s_{i-1}(a)}{w_i(a)}(q - s_{i-1}(a)) - s_{i-1}(a)p.$$

We use open gates; their boundary is the impact singular set.

Lemma 1.2 (Exact channel realization). *For each i there is an exact symplectic cobordism $(C_{a,i}, d\lambda_{a,i})$, with disjoint incoming and outgoing faces identified respectively with C_i and $B_{a,i}(C_i)$, and an autonomous Hamiltonian $H_{a,i}$ whose regular incoming-to-outgoing scattering map is $B_{a,i}$. The construction is smooth in a on every compact subinterval of $(-1, 1)$.*

Proof. Use the isotopy

$$(q, p) \mapsto (s_{i-1} + e^{t\ell_i}(q - s_{i-1}), s_{i-1} + e^{-t\ell_i}(p - s_{i-1})), \quad \ell_i = -\log w_i,$$

and compose it with the two canonical translations needed to obtain the displayed affine branch. The three stages are Hamiltonian flows generated by a linear momentum, by $-\ell_i(q - s_{i-1})(p - s_{i-1})$, and by a linear position Hamiltonian. Choose a smooth clock partition $\chi_1 + \chi_2 + \chi_3 = 1$ on $0 < t < 1$, so the stages have disjoint clock support. Adjoin a canonical clock pair (t, E) and take $H = E + \sum_k \chi_k(t)H_k(q, p)$. Then $\dot{t} = 1$, and the time-one scattering map is exactly the composition, hence $B_{a,i}$. The open gate C_i selects the channel geometrically; no external reset is applied. The primitive $G_{a,i}$ supplies the exact face matching of $\lambda = p dq + E dt - dG_{a,i}$ at the outgoing end. Smooth dependence follows from $w_i(a) > 0$. $\square$

Let $\mathcal{M}_a^\circ$ be the quotient of the disjoint channels by their outgoing-to-incoming face identifications. Only regular faces are glued; seams are retained as ideal impact faces in the graph completion.

Theorem 1.3 (Autonomous impact network with mechanical work). *The regular quotient $\mathcal{M}_a^\circ$ is a Hausdorff exact symplectic manifold carrying a complete autonomous Hamiltonian flow whose first return to the physical square S_a° is B_a . After adjoining a canonical work pair (z, π_z) , the Hamiltonian*

$$\tilde{H}_a = H_a + \pi_z c_i \quad \text{on channel } i$$

satisfies $\dot{z} = c_i$, $\dot{\pi}_z = 0$. Thus the increment of the genuine canonical coordinate z in one return is c_i , and after n returns it is the additive current A_n .

Proof. Every gluing map is the exact symplectic scattering map of Lemma 1.2; exact face matching gives a well-defined primitive on the quotient. Because the gates are disjoint open sets and the gluing is between paired regular faces, local quotient charts are ordinary mapping-torus charts. Distinct gates remain separated, so the quotient is Hausdorff. The clock vector field crosses each face transversely, and its return map is the channel scattering map. The added term is Hamiltonian on the product symplectic form $d\lambda_a + dz \wedge d\pi_z$, and Hamilton's equation gives the stated work increment. $\square$

1.3. The graded geometric flag complex. For integers $m, n \geq 0$, let $\mathcal{P}_{m,n}(a)$ be the finite polyhedral partition of the square generated by seam images through time m and seam preimages through time n . A flag $F = (F_0 \supset F_1 \supset \cdots \supset F_k)$ is an oriented chain of faces of codimension $0, \dots, k$. Let $\mathbf{M}(T)$ denote the mass of a finite current T .

For $q \geq 0$, define $\mathcal{F}_a^q$ as the completion of compatible families of currents T_F , over all finite partitions and all $k \leq q$, for the norm

$$\|T\|_{\mathcal{F}_a^q} = \sum_{k=0}^q \gamma^k \sup_{m,n} \sum_{\substack{F \in \mathcal{P}_{m,n}(a) \\ \text{codim } F=k}} \sum_{j=0}^{q-k} \mathbf{M}(D_a^j T_F), \quad \gamma > 0.$$

Compatibility means restriction under refinement and cancellation of the two opposite orientations on every internal face. There is no factor decaying with refinement depth.

Proposition 1.4 (Nondegenerate complete flag complex). *The displayed expression is a norm. Compatible refinement maps are isometries on bulk mass and contractions on total flag mass. The resulting space is Hausdorff and complete, and the constant bulk density has norm one. For each finite q , material differentiation is bounded*

$$\mathfrak{D}_a : \mathcal{F}_a^{q+1} \longrightarrow \mathcal{F}_a^q.$$

The projective intersection $\mathcal{F}_a^\infty = \bigcap_q \mathcal{F}_a^q$ supports derivatives of every finite order.

Proof. For a current restricted to a disjoint refinement, total variation is additive; hence the sum of bulk masses is independent of the partition. The same statement holds on each oriented face after internal orientations are identified. Therefore a nonzero compatible current has positive mass on some flag and cannot have zero norm. In particular the restrictions of the constant density have masses equal to the cell areas, whose sum is one.

Differentiating a moving k -face gives its material derivative on that face plus the boundary contraction with its normal velocity, which is a $(k+1)$ -flag current. Thus one derivative raises codimension by at most one and consumes one a -derivative. The mass estimate for pushforward by the affine branch and the finite seam velocity bound give the displayed bounded map. Completeness follows componentwise from completeness of finite-mass currents and the closed compatibility relations. Iterating at finite order and taking the projective intersection proves the last assertion. $\square$

1.4. Typed response. A transported observable is part of the data: it is a family $F_a \in \mathcal{F}_a^\infty$ obtained from one reference current by the declared physical transport. Let $\ell_a(F) = \int F dq dp$. The future and past Bernoulli conditional expectations have one-sided reduced resolvents $R_a^\pm$ on their centered Hölder spaces.

Theorem 1.5 (Bulk–score–flag response identity). *For every q and every C^{q+1} transported family F_a ,*

$$\frac{d^q}{da^q} \ell_a(F_a)$$

is a finite sum of words formed from: the one-sided reduced resolvents $R_a^\pm$, derivatives of the Bernoulli score, the coupled present-symbol insertion $x_0 y$, and the flag derivatives $\mathfrak{D}_a^j F_a$, with every word bounded $\mathcal{F}_a^{q+1} \rightarrow \mathbb{R}$. The formula is independent of finite cylinder presentation. If $F_a \equiv F$ is a genuinely fixed physical observable, then all score and seam terms cancel and the response is zero.

Proof. First take a finite bi-cylinder observable. Differentiate its exact rectangle integral. Interior differentiation gives the transported bulk term, differentiation of Bernoulli weights gives the centered score, and the Reynolds boundary formula gives the oriented seam current. Conditional martingale decomposition of the score into future and past coordinates produces the two one-sided Poisson resolvents; the present symbol is inserted through the coupled map of Theorem 1.1, not through two independent shifts. Repeated differentiation yields flags of codimension at most q , exactly as in Proposition 1.4.

The mass norm makes common refinement an isometry, so the formula agrees on all presentations and extends by density. For a fixed physical F , $\ell_a(F) = \int_{(0,1)^2} F dq dp$ is independent of a ; the differentiated change-of-variables identity gives the asserted cancellation. $\square$

Theorem 1.6 (Round-fourteen A1 closure). *The generalized-baker benchmark has a correctly coupled symbolic natural extension, an autonomous exact Hamiltonian impact realization with a canonical work coordinate, and a nondegenerate graded flag-current response calculus of every finite order.*

Proof. Combine Theorems 1.1, 1.3, and 1.5, with Proposition 1.4. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-fourteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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